Mode stability of subextremal Kerr–de Sitter black holes

Teukolsky ($s = \pm 2$) and ungauged linearised Einstein equations: the proof as it stands

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Abstract

For every subextremal Kerr–de Sitter (KdS) black hole we prove (A) that the spin ± 2 Teukolsky equation has no nonzero future-regular mode solutions with $\operatorname{Im}\sigma \geq 0$, apart from the allowed stationary ones, and (B) that every generalised mode solution of the ungauged linearised Einstein equation $D\mathrm{Ric}(h) = \Lambda h$ with $\operatorname{Im}\sigma \geq 0$ is pure gauge if $\sigma \neq 0$, and a linearised KdS metric plus pure gauge if $\sigma = 0$. Statement (B) is the hypothesis of the full-subextremal nonlinear stability theorem announced by Hintz–Petersen–Vasy. **Status.** Part (A) is proved for all real σ; for $\operatorname{Im}\sigma > 0$ it follows by continuity along a roots-scaling path to Schwarzschild–de Sitter, *conditional* on one analytic input (high-frequency control of QNMs in $\operatorname{Im}\sigma \geq 0$, Section 4). Part (B) is proved given (A); in the large-A regime R3 of the event lemma the multiplier is constructed by saturating one condition and solving the other with equality (Section 2.3), following the structure of Hintz's Klein–Gordon argument. Every step marked *[cert]* uses an exact symbolic or a short numerical verification; the scripts are named in the repository ledger `kds/LEMMAS.md` (item numbers below).

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1 Setting and statements

KdS with parameters (Λ, M, a), $\Lambda > 0$, in Boyer–Lindquist coordinates has $\Delta_r = (r^2 + a^2)(1 - \Lambda r^2/3) - 2Mr = -\frac{\Lambda}{3}\prod_{j=0}^{3}(r - r_j)$, $\alpha = \Lambda a^2/3$, $\Xi = 1 + \alpha$. *Subextremal* means four distinct real roots $r_3 < 0 < r_0 < r_1 = r_e < r_2 = r_c$; then $\sum r_j = 0$. Modes are $e^{-i\sigma t + im\varphi}$ times functions of (r, θ), smooth across both horizons in the regular time t_*.

Theorem 1 (A). *For every subextremal KdS the spin $s = \pm 2$ Teukolsky equation has no nonzero future-regular mode with $\operatorname{Im}\sigma \geq 0$, except those with $\sigma = 0$ permitted below.*

Theorem 2 (B). *Let h be a generalised mode, $(L_T + i\sigma)^k h = 0$ with $\operatorname{Im}\sigma \geq 0$, of $D\mathrm{Ric}(h) = \Lambda h$, smooth across both horizons. If $\sigma \neq 0$ then $h = \mathcal{L}_X g$ with X smooth; if $\sigma = 0$ then $h = c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$ with X polynomial in t_*.*

2 Part A: Teukolsky mode stability

2.1 Heun form and the window

The Möbius map $r_0, r_1, r_2, r_3 \mapsto 0, 1, t, \infty$ puts the radial equation in Heun form with $\mu(z) = z(z - 1)(t - z)$, $t = \frac{(r_1 - r_3)(r_2 - r_0)}{(r_1 - r_0)(r_2 - r_3)} > 1$ (item 26). With $K(r) = \omega(r^2 + a^2) - am$ and $\kappa_j = i\Xi K(r_j)/\Delta_r'(r_j)$, put $e_1 = 2(\kappa_1 + \kappa_3)$, $e_t = 2(\kappa_2 + \kappa_3)$. For real ω the dual equation is real and the Wronskian identity gives $\operatorname{Im} e_1 \, |c_1|^2 = t \operatorname{Im} e_t \, |c_t|^2$ (item 34).

Proposition 3 (the window; item 88). *Let $m \neq 0$, $w = \omega/m$, $\rho_j = r_j^2 + a^2$, $\Omega_j = a/\rho_j$, $D_j = |\Delta_r'(r_j)|$, $P_j = \rho_j/D_j$. Then $\operatorname{Im} e_1 = 2\Xi m f_1(w)$, $\operatorname{Im} e_t = 2\Xi m f_2(w)$ with $f_1 = P_1(w - \Omega_1) + P_3(w - \Omega_3)$ increasing and $f_2 = -P_2(w - \Omega_2) + P_3(w - \Omega_3)$ decreasing; their roots satisfy $0 < w_2 < w_1$. Hence $\operatorname{Im} e_1 \cdot \operatorname{Im} e_t > 0$ exactly on the open window (w_2, w_1).*

Proof. $\operatorname{sign}\Delta_r'(r_j) = (-, +, -, +)$. Since ρ/Δ_r and $1/\Delta_r$ have zero residue at infinity, $P_1 + P_3 = P_0 + P_2$ and $D_1^{-1} + D_3^{-1} = D_0^{-1} + D_2^{-1}$. So the slope of f_2 is $P_0 - P_1$, and $P_1 > P_0$ because $D_0/D_1 = f(r_0)/f(r_1) > 1$ for the concave parabola $f(x) = (x - r_3)(r_2 - x)$ with vertex $-(r_0 + r_1)/2 < r_0$, while $\rho_1 > \rho_0$. This also gives $w_2 = a(D_1^{-1} - D_0^{-1})/(P_1 - P_0) > 0$. Finally, cross-multiplying $w_2 < w_1$, the a^2-terms cancel and $D_0 D_1 D_3$ times the difference factors as $2(r_0 - r_1)(r_0 - r_3)(r_1 - r_3)(r_0 r_3 - r_1 r_2) > 0$. □

Corollary 4. *(i) Outside the window, and for $m = 0$, $\omega \neq 0$ real, there is no real-frequency mode. (ii) $\omega = 0$ lies outside the window for $m \neq 0$. (iii) At w_2 only $\operatorname{Im} e_t$ vanishes and at w_1 only $\operatorname{Im} e_1$; both endpoints are then closed by the Wronskian and the injectivity of the transform (item 44).*

2.2 Inside the window: transform, dual potential, event lemma

Inside the window the Riemann–Liouville transform $I_1^{\mu+1}$ maps a quasinormal mode injectively to a solution of a dual Heun equation with the correct boundary branches (items 17, 55). The dual potential is exactly

$$L(z) = \frac{A}{z} + B - 3z, \qquad A = t\big(X_0^2 - (X_1 + X_t)^2\big) > 0, \qquad B = \lambda C_0 + c_1,$$

with $C_0 = 3/(\Lambda(r_1 - r_0)(r_2 - r_3))$ and $c_1 = 1 + t + 3(1 - \alpha)C_0$.

Lemma 5 (the c_1 identity; item 89, *cert*). *With $\Lambda = 3$ (by scaling) and the exact relations $\sum \kappa_j = 0$, $\sum r_j^2 \kappa_j = 0$, $\sum r_j \kappa_j = -i\Xi\omega$ (residue identities), as rational functions $n_1 - (\lambda C_0 + c_1) = -2(s^2 - 4)\frac{r_0^2 + r_0 r_1 + r_0 r_2 - r_1 r_2}{(r_0 - r_1)(r_0 + r_1 + 2r_2)}$ and $n_2 + 3 = 4 - s^2$.*

The angular bound $\lambda \geq 3/2$ on the window (item 46) and $C_0 > (2t-1)/(3(1-\alpha))$ (item 72, exact form $C_0 - \frac{3}{2} \cdot \frac{2t-1}{3(1-\alpha)} = \frac{3(r_0+r_1)^2}{2\Lambda(r_1-r_0)(r_2-r_3)}$) give the floor $B \geq 5.5t - 1.25$.

The two-current argument (items 8, 12, 17) reduces mode stability inside the window to the following.

Definition 6 (event lemma). For $t > 1$, $z_0 \in (1, t)$, $A > 0$, $B \geq 5.5t - 1.25$, find h on $(1, z_0)$ with $s = z_0 - z$, $H = 2\mu + s\mu'$ and

$$\text{(C)} \quad s\,h\,\mu \leq H, \qquad \text{(D)} \quad \tfrac{1}{2}\big(\mu' h + \mu(h' + h^2)\big) \leq s(Lh + L').$$

Regimes R1 ($p \leq 1$) and R2 ($p > 1$, $2(z_0-1)^2(A/z_0+3) < t-1$), with $p = 2(z_0-1)L(1)/(t-1)$, are closed by the explicit ODE multiplier of item 47. In R3 we use the rational multiplier (item 104)

$$h = 1.7\,g\,\frac{w}{1+w} + \frac{0.45\,(1-u)}{xz}, \quad g = -\frac{L'}{L},\ w = 8\kappa s^2,\ \kappa = \frac{L(z_0)}{\mu(z_0)},\ x = z - 1,\ u = \frac{x}{z_0 - 1}.$$

Proposition 7 ((C) in R3; item 105). *(C) holds for all t, z_0, A and $B \geq 5.5t - 1.25$.*

Proof. All three factors of h are nonnegative, $w/(1+w) < 1$, $1 - u \leq 1$, so it suffices that $1.7sg + 0.45s/(xz) \leq 2 + s/z + s/x - s/(t-z)$ (divide by μ). Now $g - 1/z = (6 - B/z)/L$, so $g \leq 1/z$ when $B \geq 6z$ and otherwise the excess is at most $(0.5 + 1.25/t)/(2.5t - 1.25)$; $0.7s/z + 0.45s/(xz) - s/x \leq -0.3s/z$; $s/(t-z) < 1$; and where the excess is nonzero $1.7s \cdot \text{excess} \leq 0.69$. Summing, the difference is at most $-0.3s/z + 1 + 0.69 - 2 < 0$. □

2.3 Regime R3: saturate (C), solve (D)

Write $h = f'/f$. Then (D) $\iff \widetilde{D}f := (\mu f')' - 2s(Lf)' \leq 0$ and (C) $\iff h \leq h_C := H/(s\mu) = 2/s + \mu'/\mu$.

Proposition 8 (items 117–121). *For every R3 parameter point there is a C^2 function $f > 0$ on $(1, z_0]$ with $f(1) = 0$, (C) and (D) on $(1, z_0)$, and $f'(z_0) > 0$.*

Proof. Construction. On $(1, \hat{z}]$ let $f = F_C = \mu/s^2$, so (C) holds with equality; on $[\hat{z}, z_0]$ let f solve $\widetilde{D}f = 0$ with matching data, so (D) holds with equality. Here $\hat{z}$ is the unique zero of $d := \widetilde{D}F_C/(2F_C) = N(S)/(2F_C s^4)$, $S = z_0 - z$.

(P1): d has exactly one sign change. N is a sextic in S with $N(0) = 6\mu_0^2 > 0$ and vanishing S^4 coefficient. After $S = x_0 v/(1+v)$ its seven coefficients $T_0, \ldots, T_6$ satisfy $T_0, T_1 > 0$; $T_2, \ldots, T_6$ are affine in (A, B) with negative coefficients; the feasible (A, B)-set is the quadrant $\{A \geq \max(0, A_{c2}), B \geq 5.5t - 1.25\}$; and on it $T_4, T_5, T_6 < 0$ and $T_3 \geq 0 \Rightarrow T_2 > 0$ (two-variable statements, exact Bernstein certificate). Descartes' rule gives exactly one root.

(C) on the second piece. With $p = \mu \exp(-\int 2sL/\mu)$, the Wronskian $W = p(F_C f' - F_C' f)$ satisfies $W' = -(p/\mu) f\, \widetilde{D}F_C \leq 0$ and $W(\hat{z}) = 0$, so $f'/f \leq h_C$.

(P2): positivity and the seam. At a zero of f' with $f > 0$, $\mu f'' = 2sL'f < 0$, so $f'(z_0) > 0$ implies $f' > 0$ and $f > 0$ throughout. With $\Phi = \mu f' - 2sLf$, $\Phi' = 2Lf$; a first zero of f' would force $\Psi(z_1) \leq 0$ for a function Ψ with $\Psi' < 0$ and $\Psi(z_0) = \hat{H}/\hat{s} - 2\int_{\hat{z}}^{z_0} s|L'|$, and $\int_z^{z_0} s|L'| \leq s^2(A/(2z^2) + 3/2)$ reduces $\Psi(z_0) > 0$ to $K(\hat{z}) > 0$, $K = H - s^3|L'|$. Since K and N are affine decreasing in A (and N in B), $K \leq 0 \Rightarrow N(S; \max(A_K, 0), 5.5t - 1.25) < 0$, a three-variable statement certified exactly; as $N \geq 0$ on $[0, \hat{S}]$, $K(\hat{z}) > 0$. □

2.4 $\sigma = 0$ and the upper half plane

For $m = 0$, $\sigma = 0$: an energy identity with $V = -2\Lambda r^2 - 2(1-\alpha) - \lambda < 0$ for $s = -2$, and for $s = +2$ the Teukolsky–Starobinsky map, whose kernel is excluded by $\det = [\lambda^2 + 6(1-\alpha)\lambda + 8 - 4\alpha + 8\alpha^2]^2 > 0$ (items 35, 43). Modes with $\operatorname{Im}\sigma > 0$ are excluded by continuity from Schwarzschild–de Sitter: QNMs depend continuously on (Λ, a) (Petersen–Vasy, full subextremal Fredholm theory) and none can cross the real axis by the above (item 36).

3 Part B: ungauged linearised Einstein mode stability

Throughout, $a \neq 0$; for $a = 0$ Theorem B is Hintz–Vasy 2018. By Theorem A, $\delta\Psi_0 = \delta\Psi_4 = 0$ for every generalised mode (item B.1). We use GHP with $\text{þ} = D - p\epsilon - q\bar{\epsilon}$, $\text{ð} = \delta - p\beta - q\bar{\alpha}$, $\text{þ}' = \Delta - p\gamma - q\bar{\gamma}$, $\text{ð}' = \bar{\delta} - p\alpha - q\bar{\beta}$, $\rho' = -\mu$, $\tau' = -\pi$ (conventions fixed by machine checks on KdS, item 92). On KdS, $\Psi_2 = -M/(r - ia\cos\theta)^3$; put $\phi = r - ia\cos\theta \propto \Psi_2^{-1/3}$.

3.1 Linearised Goldberg–Sachs

Lemma 9 (item 90). *Infinitesimal null rotations achieve $\delta\Psi_1 = \delta\Psi_3 = 0$; then $\delta\kappa = \delta\sigma = \delta\kappa' = \delta\sigma' = 0$.*

Proof. To first order a null rotation about n with parameter d shifts $\delta\Psi_1$ by $3d\Psi_2$ and fixes $\delta\Psi_0, \delta\Psi_2$; about l similarly for $\delta\Psi_3$. As $\Psi_2 \neq 0$ the choice is algebraic. The vacuum-Λ Bianchi identities have no Λ terms; linearising $\text{þ}\psi_1 - \text{ð}'\psi_0 = -\tau'\psi_0 + 4\rho\psi_1 - 3\kappa\psi_2$ gives $\delta\kappa = 0$, and $\text{þ}'\psi_0 - \text{ð}\psi_1 = \rho'\psi_0 - 4\tau\psi_1 + 3\sigma\psi_2$ gives $\delta\sigma = 0$; primes give the rest. □

3.2 The Killing-spinor field and reality

On a type D Λ-vacuum (with $\kappa = \sigma = \kappa' = \sigma' = 0$) the GHP derivatives of the tetrad are as listed in item 93(b) and the following hold (item 91; machine-checked on KdS and on the accelerating NUT Plebański–Demiański metric, items 92, 97): $\text{þ}\rho = \rho^2$, $\text{ð}\rho = (\rho - \bar{\rho})\tau$, $\text{þ}\tau = (\tau - \bar{\tau}')\rho$, $\text{ð}\tau = \tau^2$ and primes, $\text{þ}\rho' - \text{þ}'\rho = \tau\bar{\tau} - \tau'\bar{\tau}'$, $\text{ð}\tau' - \text{ð}'\tau = \rho\bar{\rho}' - \bar{\rho}\rho'$, $\text{þ}\tau' = \text{ð}'\rho$, $\text{þ}'\tau = \text{ð}\rho'$.

Lemma 10 (items 93, 96). *With $A = \phi\rho$, $B = -\phi\rho'$, $C = \phi\tau$, $D = -\phi\tau'$, the field $\xi = Bl + An - Dm - C\bar{m}$ satisfies all ten Killing equations as an algebraic consequence of these identities. On KdS, $\xi = -\Xi\,\partial_t$.*

Hence for a perturbation in the gauge above, $\mathcal{L}_\xi h + \mathcal{L}_{\delta\xi} g = 0$ with $\delta\xi$ a local linear expression in h commuting with $\mathcal{L}_{\partial_t}, \mathcal{L}_{\partial_\varphi}$.

Lemma 11 (reality for modes; item 96). *Let $J[h] = (\delta\xi[h] - \overline{\delta\xi[\bar{h}]})/(2i)$. Then $\mathcal{L}_{J[h]} g = 0$, so $J[h]$ is a constant combination of $\partial_t, \partial_\varphi$ (the local Killing algebra of KdS with $a \neq 0$ is two-dimensional, item 97 R2). For a mode with $(\sigma, m) \neq (0, 0)$, $J[h]$ is a mode of the same type and so vanishes.*

Consequently, to first order, $\phi\rho = \bar{\phi}\bar{\rho}$, $\phi\rho' = \bar{\phi}\bar{\rho}'$, $\phi\tau = -\bar{\phi}\bar{\tau}'$.

3.3 The second Killing field

Put $R = \operatorname{Re}\phi$, $I = \operatorname{Im}\phi$ (the Carter coordinates r, $a\cos\theta$). Reality gives $\text{þ}R = -A$, $\text{þ}'R = B$, $\text{ð}R = \text{ð}'R = 0$, $\text{þ}I = \text{þ}'I = 0$, $\text{ð}I = iC$, $\text{ð}'I = -iD$.

Lemma 12 (items 96, 97). *$\zeta = I^2(Bl + An) + R^2(Dm + C\bar{m})$ satisfies all ten Killing equations by algebra (the ln, $m\bar{m}$, lm, nm components are computed in item 96–97 and cancel by reality; the others are conjugates). On KdS $\zeta = -\Xi(a^2\partial_t + a\partial_\varphi)$.*

3.4 Conclusion for $(\sigma, m) \neq (0, 0)$

From $\mathcal{L}_\xi h = -\mathcal{L}_{\delta\xi} g$ and $\mathcal{L}_\zeta h = -\mathcal{L}_{\delta\zeta} g$: if $\sigma \neq 0$, $h = \mathcal{L}_X g$ with $X = \delta\xi/(i\sigma\Xi)$; if $\sigma = 0$, $m \neq 0$, $h = \mathcal{L}_X g$ with $X = -\delta\zeta/(im\Xi a)$. X is smooth across the horizons because $\xi, \zeta, \delta\xi, \delta\zeta$ are boost-spin invariant natural constructions from the Weyl spinor (item 97 R3). HPV's modes with zero T-frequency and $m \neq 0$ have ∂_t-frequency $\Omega m \neq 0$.

3.5 $\sigma = m = 0$

Proposition 13 (items 98, 102, 103)**.** *A stationary axisymmetric h with $\delta\Psi_0 = \delta\Psi_4 = 0$ is $c_M \partial_M g + c_a \partial_a g + \mathcal{L}_X g$.*

Sketch. The constants in $J[h]$ are removed by a phase of ϕ and by the acceleration tangent, which is the only Plebański–Demiański tangent making ξ complex (item 95, exact nonlinear check). Then ξ, ζ are first-order Killing fields, they commute, $\nabla R \perp \nabla I$, $\Delta_r := -2AB|\phi|^2$ depends on R only and $\Delta_x := -2CD|\phi|^2$ on I only, and in adapted coordinates the perturbed metric is Carter's ansatz to first order. On that ansatz the Λ-vacuum equations are linear in (Δ_r, Δ_x), with solutions exactly $\Delta_r = -\frac{\Lambda}{3}R^4 + \varepsilon R^2 - 2mR + \gamma$, $\Delta_x = -\frac{\Lambda}{3}I^4 - \varepsilon I^2 + 2nI + \gamma$ (item 103, *cert*). Modulo the scaling, the variations are $\delta M, \delta a, \delta n$; with acceleration, four. Axis regularity (determinant $-64mp_0^4/c \neq 0$) removes NUT and acceleration. □

3.6 Generalised modes

For $(\sigma, m) \neq (0, 0)$, induction on k: the top coefficient is a mode, hence $\mathcal{L}_{X_1} g$, and $Y = t_*^{k-1} X_1/(k-1)!$ lowers the order. For $\sigma = m = 0$, linear growth of $(\delta M, \delta a)$ is excluded by the Iyer–Wald charges: they are surface-independent, vanish on every $\mathcal{L}_X g$ by covariance of the Noether charge (item 101), and $\partial(E, J)/\partial(M, a)$ has determinant $M/\Xi^4 \neq 0$ for $E = M/\Xi^2$, $J = aM/\Xi^2$ (item 100).

4 What remains

1. **Analytic input for $\operatorname{Im}\sigma > 0$ in (A).** The continuity argument needs that no $s = \pm 2$ Teukolsky QNM enters $\operatorname{Im}\sigma \geq 0$ from infinity, locally uniformly in the parameters. Discreteness and continuity follow from the Vasy framework (Petersen–Vasy set up the Teukolsky Fredholm theory); the missing piece is a subprincipal-symbol bound at the trapped set for Teukolsky on full-range KdS. Hintz–Petersen–Vasy (arXiv:2508.06620, Thm. 3.1) prove such a bound for the gauge-fixed linearised Einstein operator; either verifying it for Teukolsky or running the continuity argument on their gauged operator with their constraint-damping transfer (Thm. 2.7) closes it.

2. The Pochhammer write-up of the item-17 transform and the cited bound $\alpha \leq 7 - 4\sqrt{3}$.

3. An independent check of the whole argument, in particular of Part B (the in-house hostile passes are items 97 and 102) and of the R3 construction (items 117–121).